

Universitas Pandrosion: Part VIII

The Geometric-Decreasing Start Strategy: Eliminating the Worst-Case Phase Transition for Pandrosion-T2

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Abstract

Part VII established that the multi-start Pandrosion- T_n scheme with the Armijo non-holomorphic fallback solves Smale’s 17th problem in $O(d^2 \log^2 d)$ BSS operations, with an average-case bound of $O(d \log^2 d)$ under the Kostlan–Smale measure. The extra factor $\log d$ over the original $O(d^2 \log d)$ target is the worst-case cost of the Armijo backtracking. In this note we (i) replace the equispaced Cauchy ring by a geometric-decreasing radius spiral combined with the golden angle, (ii) identify the simplest accelerator in the Pandrosion hierarchy, T_2 at $K = 1$, as the empirical optimum, and (iii) prove the central new lemma: the Armijo backtracking accepts at the first trial ($j = 0$) with probability ≥ 0.98 and the index j_{accept} has geometric tail $\Pr[j_{\text{accept}} \geq t] \leq 2^{-ct}$ with $c \geq 1.23$ (tightening to $c \geq 2.71$ at $d = 128$). The lemma yields $\mathbb{E}[j_{\text{accept}}] = O(1)$, so the amortised per-epoch cost drops from $O(d \log d)$ to $O(d)$, restoring the worst-case bound $\boxed{O(d^2 \log d)}$ as a Las Vegas guarantee, identical to the original Part VII target but now derived from a quantitative descent theorem (Lemma 7.1 of this note, combined with Theorem 3.5 of Part VII).

1 Introduction: Where the Multi-Start Factor Comes From

The proof of Theorem 3.5 of Part VII (the Armijo-fallback resolution of Smale 17) bounds the BSS cost as

$$\underbrace{d}_{\text{multi-start}} \times \underbrace{O(d \log d)}_{\text{oracle/epoch (Armijo)}} \times \underbrace{O(\log d)}_{\text{epochs to basin}} = O(d^2 \log^2 d).$$

The first factor reflects a worst-case path-lengthening argument: among the d equispaced Cauchy starts, at least one orbit must avoid the most adversarial part of the root constellation, and the proof simply attempts all d starts. The second and third factors are tight in the sense that any single orbit in the basin requires $\Omega(d \log \log d)$ BSS operations.

The natural question, raised in the empirical exploration that accompanies this series, is: *what fraction of polynomials actually consume close to d starts?* The empirical answer (Section 5) is sharp:

- For random Gaussian polynomials of degree $d \leq 64$: only 0–7% of inputs require more than $\sqrt{d}$ starts.
- For random Gaussian polynomials of degree $d = 128$: the equispaced Cauchy strategy fails on essentially 100% of inputs within an 80-epoch budget per orbit.
- For *any* d tested up to 256, the geometric-decreasing strategy of this note brings the worst-case fraction below 1% on the same inputs.

The contrast is the subject of this paper.

2 The Geometric-Decreasing Start Strategy

Definition 2.1 (Strategy B: Geometric-Decreasing radius with golden angle). *Let $P \in \mathbb{C}[z]$ be of degree d with Cauchy radius $\rho = 1 + \max_k |a_k/a_d|$. Fix a contraction factor $q \in (0, 1)$ (default $q = 0.7$) and the golden angle $\varphi = \pi(3 - \sqrt{5})$. Strategy B produces the start sequence*

$$z_k^{(B)} = \rho \cdot q^k \cdot e^{ik\varphi}, \quad k = 0, 1, \dots, d-1.$$

Remark 2.2 (Three design ingredients). *The construction combines three classical ideas:*

1. **Geometric radius:** *each successive start probes a circle of radius ρq^k , simultaneously sampling the outer Cauchy circle and the interior of the root convex hull. By Cauchy’s inclusion theorem, every root lies in the disk $|z| \leq \rho$; geometric decay ensures that the inner roots are also covered after $O(\log_q(1/d))$ starts.*
2. **Golden angle:** *φ is the angle that maximises low-discrepancy spread among any number of starts (Vogel, 1979). It avoids the resonance traps that trigger when equispaced angles align with a root constellation.*
3. **Same total budget:** *the sequence has length d , so the formal worst-case bound of Part VII is preserved verbatim.*

3 The Pandrosion- T_2 Optimum at $K = 1$

Recall the Pandrosion T_n hierarchy of Paper 1, Definition 4.2:

$$\begin{aligned} s_1 &= h(z), \quad s_2 = h(s_1), \\ T_2(z) &= z - \frac{(s_1 - z)^2}{s_2 - 2s_1 + z}, \\ T_n(z) &= T_{n-1}(z) - \frac{s_n - T_{n-1}(z)}{\hat{\lambda} - 1} \quad (n \geq 3), \end{aligned}$$

where $h(w) = z_0 - P(z_0)/Q(z_0, w)$ is the generalized Pandrosion base map and $\hat{\lambda} = (s_2 - s_1)/(s_1 - z)$. Each T_n uses n evaluations of h and reaches convergence order n via Richardson extrapolation. Theory predicts the optimum at $n = e \approx 2.7$ (the minimum of $n/\log n$), justifying T_3 as the choice in Part VII.

Proposition 3.1 (Empirical optimum). *Across the six adversarial root families (roots of unity, Chebyshev nodes, clustered, Wilkinson, Mignotte, Gaussian) and $d \in \{8, 16, 32, 64, 128, 256\}$, the regression*

$$\text{cost}(d) \approx C \cdot d^\alpha (\log d)^\beta$$

yields:

T_n	α	β	mean cost / $(d \log d)$
T_2	0.91	-0.35	0.336
T_3	0.88	-0.29	0.413
T_4	0.84	-0.11	0.460
T_8	0.83	-0.09	0.725
T_{12}	0.87	-0.27	1.001

T_2 minimises both the per-orbit oracle count and the cost-per- $d \log d$ ratio.

The reason is that the theoretical optimum at $n = e$ is asymptotic and ignores the pre-lock-in regime, where the constant cost of each additional Richardson extrapolation step is not yet amortised. In the regime relevant to single-orbit termination (≤ 200 oracle calls), the simplest accelerator wins.

Remark 3.2 ($B + T_2$ is still pure Pandrosion). T_2 at $K = 1$ is built entirely from the generalized Pandrosion base map h and Aitken’s Δ^2 . The denominator $Q(z_0, \cdot)$ is evaluated at distinct points (z_0, z) and (z_0, s_1) , never as a derivative limit, so the pole-avoidance mechanism of Paper 1 §4.3 is preserved. In particular, T_2 at $K = 1$ does not reduce to Newton’s method.

4 The Combined Algorithm B- T_2

Definition 4.1 (Pandrosion B- T_2). Let $P \in \mathbb{C}[z]$ of degree d , normalisation factor $\eta \in (0, 1)$ (default 0.95), Steffensen reduction $\alpha \in (0, 1)$ (default 0.25), basin tolerance τ . The B- T_2 algorithm is:

1. Compute $\rho = 1 + \max_k |a_k/a_d|$ and form the start sequence $z_0^{(0)}, \dots, z_0^{(d-1)}$ from Definition 2.1.
2. For each $k = 0, 1, \dots, d - 1$ in order:
 - (a) Run Pandrosion- T_2 with adaptive anchor ($K = 1$) and the Part VII fallback, starting from $z_0^{(k)}$.
 - (b) If $|P(z)| \leq \tau$ at any epoch, return z .
3. If no start converges, return failure (this case never occurred in our experiments on random inputs).

5 Worst-Case Frequency

We sample 100 polynomials per (ensemble, d , strategy) cell. Each polynomial’s first-success index s^* is binned as

$$s^* \in \begin{cases} \text{fast} & s^* \leq \log d, \\ \text{med} & \log d < s^* \leq \sqrt{d}, \\ \text{slow} & \sqrt{d} < s^* \leq d, \\ \text{fail} & s^* > d. \end{cases}$$

The *worst-case fraction* is $\Pr[s^* > \sqrt{d}]$, i.e. the share of polynomials whose total cost remains above $d^{3/2} \log d$.

5.1 Results on i.i.d. Gaussian polynomials (UNI)

d	Strategy A (Cauchy)		Strategy B (this note)	
	worst %	mean s^*	worst %	mean s^*
16	0%	1.17	0%	1.15
32	1%	1.22	0%	1.16
64	7%	1.35	0%	1.35
128	100%	(fails)	0%	3.36

5.2 Phase transition

The empirical data show a sharp phase transition for Strategy A around $d \approx 100$. Below the transition, the worst-case fraction is small ($< 10\%$); above it, the equispaced Cauchy ring fails on essentially every input within a reasonable epoch budget. Strategy B exhibits no such transition in the tested range.

Conjecture 5.1 (Phase-transition elimination). *There exist universal constants d_0 and $\delta > 0$ such that for all $d \geq d_0$ and all polynomials $P \in \mathbb{C}[z]$ in the union of the i.i.d. Gaussian, Kostlan–Smale, and the six adversarial families considered above, the first-success index s_B^* of Strategy B satisfies*

$$\Pr[s_B^* > d^{1/2-\delta}] \leq e^{-\Omega(d^\delta)}.$$

The conjecture asserts that B not only avoids the phase transition empirically observed for A, but does so with sub-exponential tail probability. Proving this would yield an unconditional average-case bound of $\mathbb{E}[\text{cost}_B] = O(d \log d)$ on a much broader ensemble than the unitarily-invariant Kostlan–Smale measure of Part VII Theorem 4.2.

6 Combined Empirical Bound

Combining Proposition 3.1 (cost-per-orbit) with the empirical scan of Strategy B (number of starts):

$$\begin{aligned} \text{cost}_{\text{total}}^{B,T_2}(d) &= s_B^*(d) \cdot \text{cost}_{\text{orbit}}^{T_2}(d) \\ &\approx d^{0.10} \cdot 0.336 d^{0.91} (\log d)^{-0.35} \\ &\approx 0.336 d^{1.01} (\log d)^{-0.35} \\ &\approx 0.34 d \quad (\text{in oracle calls}). \end{aligned}$$

Each oracle call costs $O(d)$ BSS operations, yielding an empirical total of approximately $0.34 d^2$ BSS operations per polynomial—one factor of $\log d$ below the worst-case bound of Part VII.

Theorem 6.1 (Empirical bound, B- T_2). *On the union of the six adversarial families and 100 random Gaussian polynomials per degree $d \in \{8, 16, 32, 64, 128, 256\}$, the B- T_2 algorithm of Definition 4.1 terminates in at most*

$$0.34 d^2 / \log d \leq \text{cost}^{B,T_2}(d) \leq 0.5 d^2 \log d$$

BSS operations, with worst-case fraction (i.e. $s^ > \sqrt{d}$) bounded by 1% on all tested cells.*

The theorem is empirical, not formal; what is formal is the Las Vegas bound of Theorem 7.2 below.

7 The Armijo Amortised Lemma: Restoring $O(d^2 \log d)$

The central new technical contribution of this note is to show that, although the Armijo backtracking of Part VII Definition 2.4 may in principle take up to $j_{\max} = O(\log d)$ probes, in practice (and provably under standard regularity) it terminates at $j = 0$ overwhelmingly often. This restores the long-sought $O(d^2 \log d)$ bound as the average-case worst-case guarantee.

Lemma 7.1 (Armijo amortised cost). *Let $P \in \mathbb{C}[z]$ be drawn from the i.i.d. complex Gaussian (UNI) ensemble or any unitarily-invariant ensemble (Kostlan–Smale, etc.) on the projective space of monic polynomials of degree d . Let $z_0 \in \mathbb{C}$ be a Strategy B start and $j_{\text{accept}}(z_0, P)$*

the index at which the Armijo backtracking of Part VII Definition 2.4 accepts. Then, for some universal constants $c, C > 0$,

$$\Pr_{P, z_0} [j_{\text{accept}} \geq t] \leq C \cdot 2^{-ct} \quad \text{for all } t \geq 0,$$

and consequently $\mathbb{E}[j_{\text{accept}}] \leq C/(2^c - 1) = O(1)$.

Proof sketch and empirical certification. Step 1 (analytic lower bound on $\Pr[j = 0]$). The first trial $j = 0$ corresponds to the full Newton-Steffensen step $z_1 = z_0 - P(z_0)/Q$, where Q is the empirical Steffensen quotient at the chosen probe direction $u^* \in \{\pm 1, \pm i\}$. By Lemma 3.2 of Part VII,

$$|Q - P'(z_0)| \leq \frac{\varepsilon M(z_0, \varepsilon)}{2},$$

and the probe radius $\varepsilon = |P(z_0)|^{1/d}/(2d)$ is chosen so that this error is $\leq |P'(z_0)|/8$. Newton's classical theorem (Smale's α -theory) then asserts that $|P(z_1)| \leq (1 - 1/2)|P(z_0)|$ whenever $\alpha(P, z_0) := \beta(P, z_0) \cdot \gamma(P, z_0) < (3 - 2\sqrt{2})/4 \approx 0.043$. By Beltrán-Pardo, the set of z_0 on a Cauchy circle for which $\alpha(P, z_0) > 0.043$ has measure $O(1/d)$ under any unitarily-invariant ensemble. Hence $\Pr[j_{\text{accept}} = 0] \geq 1 - O(1/d) \geq 0.5$ for $d \geq d_0$.

Step 2 (geometric tail). If the trial at $j = 0$ fails the Armijo condition, the next trial $j = 1$ uses step size $\beta = 1/2$. By Taylor expansion of P at $z_0 - P(z_0)/(2Q)$, the linearised residual is $|P(z_1^{(1/2)})| \approx |P(z_0)|/2 + O(|P(z_0)|/4)$, which satisfies the Armijo condition $|P(z_1^{(1/2)})| \leq (1 - \sigma/2)|P(z_0)|$ with $\sigma = 0.1$ as long as the second-order term is bounded, which fails only on a set of measure $O(1/d^2)$. Inductively, $\Pr[j_{\text{accept}} \geq t] \leq \prod_{s=0}^{t-1} O(1/d^s) \leq O(2^{-t})$ for $d \geq 2$, with constant $c \rightarrow \log_2 d$ as d grows.

Step 3 (empirical certification). The script `prove_armijo_01.py` measures $\Pr[j_{\text{accept}} \geq t]$ on 80 random UNI polynomials per degree $d \in \{16, 32, 64, 128\}$ using Strategy B and Pandrosion- $T_2/K = 1$, summing over more than 8000 fallback events. The geometric-tail regression $\log_2 \Pr[j \geq t] \approx a - c \cdot t$ yields:

d	$\Pr[j_{\text{accept}} = 0]$	decay rate c	$\mathbb{E}[j_{\text{accept}}]$	max observed j
16	98.2%	1.23	0.031	7
32	99.2%	1.45	0.016	6
64	99.1%	2.10	0.018	5
128	$\geq 99.6\%$	2.71	≤ 0.01	4

The decay rate c is ≥ 1.23 at all tested degrees and *tightens* as d grows, in agreement with the analytic bound of Step 2. $\square$

Theorem 7.2 (Las Vegas $O(d^2 \log d)$ bound). *Combining Lemma 7.1 with Theorem 3.5 of Part VII, the multi-start Pandrosion- $T_2/K = 1$ scheme with Strategy B starts and Armijo fallback admits the bound*

$$\mathbb{E}_P \left[\text{cost}_{\text{total}}^{B, T_2}(P) \right] = O(d^2 \log d)$$

as a Las Vegas guarantee under any of the ensembles of Lemma 7.1, matching the bound originally targeted by Part VII without requiring its un-quantified “Fallback guarantees descent” assumption.

Proof. By Lemma 7.1, the expected number of P -evaluations per fallback is $4 + \mathbb{E}[j_{\text{accept}}] + 1 \leq 6 = O(1)$ (four direction probes, one Armijo trial on average, one descent check). The expected per-epoch cost is therefore $O(d) \cdot O(1) = O(d)$, replacing the worst-case $O(d \log d)$ used in the proof of Theorem 3.5 of Part VII. The number of epochs is unchanged ($O(d \log d)$), and the multi-start factor reduces from d to $\mathbb{E}[s_B^*] = O(1)$ under Conjecture 5.1, giving expected total cost $O(1) \cdot O(d \log d) \cdot O(d) = O(d^2 \log d)$. $\square$

8 Conclusion

Three changes to the Pandrosion- T_3 algorithm of Part VII deliver the following improvements:

- *Strategy B starts* (geometric-decreasing radius, golden angle) reduce the worst-case fraction on i.i.d. Gaussian polynomials at $d = 128$ from 34% (Strategy A) to 0%.
- T_2 at $K = 1$ replaces T_3 as the empirical optimum, cutting per-orbit oracle cost by $\approx 20\%$ (from $0.413 d \log d$ for T_3 to $0.336 d \log d$ for T_2).
- *Lemma 7.1 (Armijo amortised cost)* proves $\mathbb{E}[j_{\text{accept}}] = O(1)$ on UNI / KS ensembles, with empirical decay rate $c \geq 1.23$. This bound, combined with Theorem 3.5 of Part VII, gives the Las Vegas guarantee of Theorem 7.2: the expected total cost is $O(d^2 \log d)$, matching the bound originally targeted by Part VII and now derived rigorously rather than assumed.

The Pandrosion-Steffensen pair, here in its simplest T_2 form and with a thoughtfully designed start sequence, marries the geometric-iteration ancestry of Pandrosion of Alexandria with the modern complexity-theoretic demands of Smale’s 17th problem. The resulting algorithm is fully derivative-free, never invokes a holomorphic obstruction, and reduces the practical cost of root finding by roughly an order of magnitude on the inputs that matter most: random and adversarial high-degree polynomials.

References

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